Transformations, Compositions and Inverses: A Synthesis Review
Answer Key

Precalculus

Quick Answers

1. 12
2. 19
3. 3
4. -6
5. $x^2 + 8x + 15$
6. 15
7. -3, 3
8. (a) value = 2, —
9. 7
10. (a) simplified form = x, —
11. 3
12. (a) value = $\frac{8}{3}$, —

What is verified

12 of 15 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 8, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSF-BF.A.1c – *problems 2, 5, 7, 10*
HSF-BF.B.3 – *problems 1, 4, 8*
HSF-BF.B.4 – *problems 3, 6, 9, 11, 12*
Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

9. *If they got -1: took the minus square root, which inverts the branch the restriction removes*

1. $f(x) = x^2$, $g(x) = f(x-2) + 3$. Find $g(5)$.
Solution: The -2 is inside f, so it changes the input: $g(5) = f(3) + 3 = 9 + 3$.
$\boxed{g(5) = 12}$

2. $f(x) = 2x + 1$, $g(x) = x^2$. Find $(f \circ g)(3)$.
Solution: Inner first: $g(3) = 9$, then $f(9) = 2(9) + 1$. $\boxed{(f \circ g)(3) = 19}$

3. $f(x) = 5x - 3$. Find $f^{-1}(12)$.

Solution: From $y = 5x - 3$, $x = \dfrac{y+3}{5}$, so $f^{-1}(x) = \dfrac{x+3}{5}$ and $f^{-1}(12) = \dfrac{15}{5}$.

$$\boxed{f^{-1}(12) = 3}$$

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4. $f(x) = \sqrt{x}$, $g(x) = -2f(x+1)$. Find $g(8)$.

Solution: $g(8) = -2\sqrt{8+1} = -2\sqrt{9} = -2(3)$. $\boxed{g(8) = -6}$

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5. $f(x) = x^2 - 1$, $g(x) = x + 4$. Write $(f \circ g)(x)$.

Solution: Substitute $x + 4$ into f:

$$(x+4)^2 - 1 = x^2 + 8x + 16 - 1$$

$$\boxed{(f \circ g)(x) = x^2 + 8x + 15}$$

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6. $f(x) = \dfrac{x-7}{2}$. Write f^{-1} and find $f^{-1}(4)$.

Solution: Undo in reverse: multiply by 2, then add 7, so $f^{-1}(x) = 2x + 7$. Then $f^{-1}(4) = 8 + 7$. $\boxed{f^{-1}(4) = 15}$

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7. $f(x) = \sqrt{x-1}$, $g(x) = x^2 + 1$. Simplify $(f \circ g)(x)$ and solve $(f \circ g)(x) = 3$.

Solution: $f\big(g(x)\big) = \sqrt{(x^2+1) - 1} = \sqrt{x^2} = |x|$. The composition is the absolute-value function, not x — the square root undoes the square only for non-negative inputs. Then $|x| = 3$ gives two solutions.

$$\boxed{x = -3 \quad \text{or} \quad x = 3}$$

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8. $f(x) = x^2 - 17$, $g(x) = 3f(x-2) + 5$. **(a)** $g(6)$. **(b)** Describe the transformations.

Solution (a): $g(6) = 3f(4) + 5 = 3\,(16 - 17) + 5 = 3(-1) + 5$. $\boxed{g(6) = 2}$

Solution (b): *Instructor-judged.* A complete answer gives a horizontal shift 2 units **right**, a vertical stretch by 3, and a vertical shift 5 units **up**, with the stretch applied before the shift. Naming all three but shifting up before stretching is wrong: the 5 would be tripled. The horizontal shift may come at any point, but "left" is not acceptable.

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9. $f(x) = (x-3)^2 + 1$ on $x \geq 3$. Find $f^{-1}(17)$.

Solution: Solve $y = (x-3)^2 + 1$: $(x-3)^2 = y - 1$, so $x = 3 \pm \sqrt{y-1}$. The restriction $x \geq 3$ selects the plus sign, giving $f^{-1}(x) = 3 + \sqrt{x-1}$. Then $f^{-1}(17) = 3 + \sqrt{16} = 3 + 4$.

$$\boxed{f^{-1}(17) = 7}$$

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10. $f(x) = \dfrac{1}{x+2}$, $g(x) = \dfrac{1}{x} - 2$. **(a)** Simplify $(f \circ g)(x)$. **(b)** What it means.

Solution (a): The -2 inside g and the $+2$ inside f cancel:

$$f\big(g(x)\big) = \frac{1}{\left(\frac{1}{x} - 2\right) + 2} = \frac{1}{\frac{1}{x}}$$

and the reciprocal of $\frac{1}{x}$ is x. $\boxed{(f \circ g)(x) = x}$

Solution (b): *Instructor-judged.* Returning x means f undoes g, so the two are inverses of each other. A strong answer adds one of: the other order should also be checked, or the identity holds only where both rules are defined, so $x = 0$ and $x = -2$ are excluded. Simply restating that the expression simplifies is not an interpretation.

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11. $f(x) = 2x + 1$, $h(x) = f^{-1}(x - 4)$. Find $h(11)$.

Solution: $f^{-1}(x) = \dfrac{x-1}{2}$. The $x - 4$ is the input to f^{-1}, so $h(11) = f^{-1}(7) = \dfrac{7-1}{2}$.

$\boxed{h(11) = 3}$

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12. $f(x) = 3x - 5$, $g(x) = x^2$ on $x \geq 0$. **(a)** $(g \circ f)^{-1}(9)$. **(b)** Why the order reverses.

Solution (a): Undo the outer function first. From g: the input to g was $3x - 5$, and g is inverted on $x \geq 0$ by the positive square root, so $3x - 5 = \sqrt{9} = 3$. Then undo f: $3x = 8$.

$$\boxed{(g \circ f)^{-1}(9) = \tfrac{8}{3}}$$

Solution (b): *Instructor-judged.* Composition applies f first and g second, so undoing it must remove g first and f second — which is f^{-1} applied *after* g^{-1}, i.e. $f^{-1} \circ g^{-1}$. A concrete illustration with these rules (take the square root, then add 5 and divide by 3) is acceptable as long as the general reversal point is made. Asserting the identity with no reasoning earns no credit.